

Hardware-Aware MILP for Optimal End-to-End Efficiency in MISO WPT: Joint Optimization of Power Modes and Fault Tolerance

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Abstract—This article presents a hardware-aware convex optimization framework for a 5.64 GHz multiple-input single-output (MISO) wireless power transfer (WPT) system. In contrast to prior work that optimizes only free-space transmission efficiency by minimizing transmit radio frequency (RF) power, the proposed method directly incorporates transmitter-side hardware dissipation into the optimization objective. Using measured data from the beamforming boards under 5 and 3.3 V supply conditions, convex dissipation models are identified through constrained regression and expressed in convex piecewise linear (PL) form. Based on these models, supply voltage switching is embedded into the WPT design through mixed-integer linear programming (MILP) formulations for 2-state 5 and 3.3 V and 3-state 5 and 3.3 V OFF operation. The framework is further extended with slack-penalized feasibility relaxation, warm-start initialization via soft transition, and dead-index embedding to explicitly account for hardware faults in large-scale arrays. The proposed method is validated using a 256-element transmitting array and a single receiving antenna at a distance of 0.5 m. The implemented system achieves 1.9 W of received RF power and 780 mW of received DC power while significantly reducing transmitter-side consumed power. The results demonstrate that the proposed dissipated-power-based formulation reduces transmitter-side consumed power when supply-voltage switching and realistic hardware constraints are taken into account.

Index Terms—Convex optimization, hardware-aware efficiency optimization, radio frequency (RF), supply voltage switching, wireless power transfer (WPT).

NOMENCLATURE

M, i Number of Tx elements/Tx-element index.

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A_i Tx RF amplitude; voltage-equivalent field domain.
 ϕ_i, θ_i Tx/channel phase.
 B_i Effective element-to-Rx channel magnitude.
 G_t, G_r, d_i, λ_0 Tx/Rx directional gains, link distance, and free-space wavelength.
 P_r Fixed receiver-aware target received RF power.
 $f(A_i), g(A_i)$ Board DC-dissipation models under 5/3.3 V rails.
 $x_i, y_{data,i}, y_{fit,i}$ Measured RF amplitude, measured/fit DC consumption, and data vectors.
 y_{data}, y_{fit}
 $m_\ell^{(f)}, c_\ell^{(f)}, m_\ell^{(g)}, c_\ell^{(g)}$ Slope/intercept of PL segment ℓ for the 5/3.3 V models.
 $f_{ref}, f_{PQ,i}, f_P$ Polynomial, convex-PQ, and convex-polynomial reference models.
 $v \in \{5, 3.3\}$ Active supply-rail index.
 $A_{i,v}, A_{max,v}$ Rail-specific Tx RF amplitude/measured amplitude limit.
 $z_{i,v}, Z_{i,v}$ Binary rail selection/complement $Z_{i,v} = 1 - z_{i,v}$; both rails zero: OFF.
 $t_{i,v}, T_v^{max}, M_v$ Epigraph dissipation, its upper bound, and big- M constant.
 D Dead or nonresponsive Tx-element set.
 s, μ Received-power feasibility slack/slack-penalty coefficient.
 $\mathbf{B}_{alive}, B_{max}$ Alive-element channel vector/strongest alive-element channel.
 A_{scale}, T_{scale} Representative amplitude/hardware-dissipation scale.
 m_{max}, η Maximum PL slope/slack-penalty safety factor.
 w_i Relaxed rail preference; 0: 5 V, 1: 3.3 V.
 $h_i(A_i, w_i)$ Soft rail-interpolation objective for warm-start generation.
 Δ_i, β, ρ Rail-wise dissipation difference, sigmoid softness, and update damping.
 $\mathbf{A}^{soft}, \mathbf{w}^{soft}$ Relaxed amplitude/rail-preference solution vectors.

I. INTRODUCTION

ANTENNA-array-based wireless power transfer (WPT) has attracted growing interest as a cable-free energy

delivery method that sustains useful efficiency over relatively long ranges, enabling applications from consumer devices to distributed sensors [1], [2], [3]. Depending on the link distance, WPT is commonly categorized into near- and far-field regimes. In practice, time reversal excitation provides a simple focusing strategy by phase conjugating the measured channel so that the transmitted fields are coherently combined at the receiver. However, this approach does not explicitly optimize the transmit amplitude distribution under a received power constraint, and therefore it does not necessarily minimize the required transmit power or the transmitter-side power consumption. This limitation motivates optimization-based WPT formulations. Our previous studies [4], [5], [6] established transmission-conversion-efficiency optimization, scalable large-array implementation, and reduced-complexity transmit-mode calibration, respectively. However, these studies and related convex-based approaches [7], [8], [9] optimized or calibrated the radio frequency (RF)-domain excitation without directly embedding measured transmitter-board dissipation or discrete hardware operating states into the optimization objective. Recent studies therefore employ convex optimization with channel knowledge to synthesize amplitude and phase weights that maximize received power under array constraints, often via semidefinite relaxation (SDR), and report near-field gains that converge toward classical beamforming in the far-field [8], [9]. Additionally, in this work, the term near-field refers specifically to the radiative near-field. In this regime, the receiver is not located in the far-field of the entire array aperture, so the propagation distance, phase, and channel magnitude vary across the array elements. At the same time, the receiver can still be regarded as being in the far-field of each individual antenna element, so the element-wise radiation characteristics can be incorporated through the corresponding channel coefficients.

Nevertheless, most large-array RF-WPT optimization and hardware-demonstration studies still treat the transmitter primarily through radiated RF power or over-the-air transfer efficiency, while the measured transmitter-board dissipation and discrete hardware operating states are not directly embedded into the optimization formulation [10], [11]. As illustrated in Fig. 1, such RF-domain formulations optimize the over-the-air transmission efficiency, η_{air} , without explicitly incorporating transmitter-side board-level dissipation into the objective. Recent waveform-engineering studies have also emphasized that end-to-end WPT efficiency cannot be improved by propagation-domain beam focusing alone. In particular, multisine waveform design can exploit beamforming gain, channel frequency selectivity, and rectifier nonlinearity to improve the harvested DC power at the receiver [12]. More recent near-field RF-WPT formulations further consider nonlinear energy harvesters, showing that the efficiency is affected by the wireless channel and receiver blocks together [13]. These studies indicate an important trend toward hardware-aware WPT design beyond a purely over-the-air RF-link viewpoint.

To mitigate this limitation, our previous work [14] heuristically applied supply-voltage switching, motivated by the strong amplitude tapering observed in convex-optimized

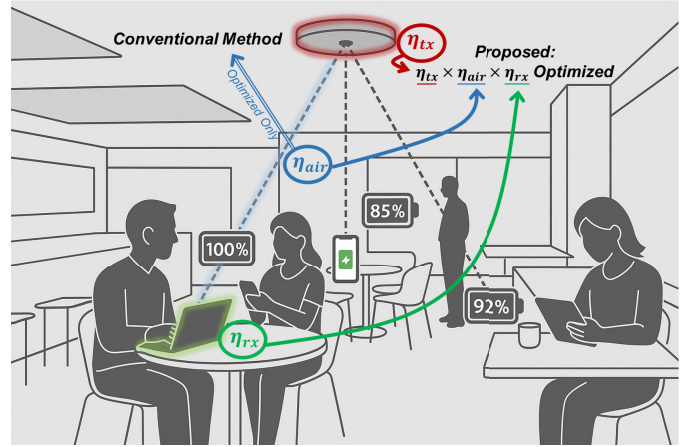


Fig. 1. Conceptual diagram of WPT system with the optimization scopes for the proposed and prior works.

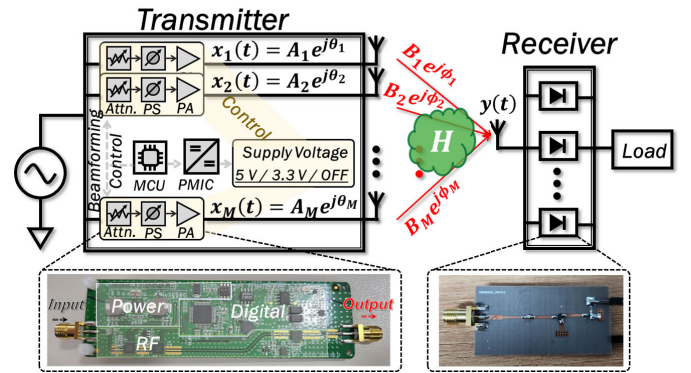


Fig. 2. System schematic representing the block diagram of the transmitter and receiver for the proposed MISO WPT.

output-power patterns. Although that approach significantly reduced DC power consumption required to generate RF power, it did not provide an optimization framework explicitly aware of hardware dissipation. In that approach, the supply state was selected heuristically after the RF excitation pattern was determined, whereas the present work jointly optimizes the excitation amplitudes and hardware states using a measured board-level dissipation objective. To overcome this gap, this work develops a hardware-aware optimization framework for a multiple-input single-output (MISO) WPT system that directly incorporates transmitter-side hardware dissipation and accounts for receiver characteristics through rectifier-aware target received-power selection. Unlike prior system-level models that typically rely on analytical component models for HPA or rectifier nonlinearities, the proposed formulation uses measured beamforming-board data to construct convex dissipation models for the actual transmitter hardware.

II. DEVELOPMENT OF THE PROPOSED WPT SYSTEM

A. Problem Formulation

As summarized in Fig. 2, the proposed MISO WPT system and its convex base formulation are as follows. The transmitter employs capacitively fed wideband antenna elements [15],

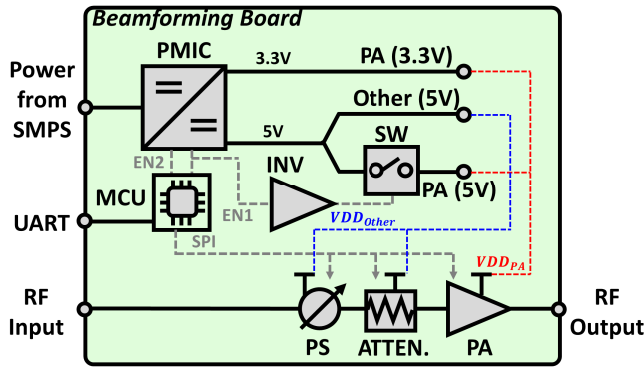


Fig. 3. Circuit-level supply and control architecture of one beamforming-board channel, including the dual-channel PMIC, EN1/EN2 control, 5 V non-PA supply path, PA supply selection, and board-level OFF operation.

[16], [17], [18], [19], [20], whose detailed design is reported in [5]. The element impedance bandwidth extends from 4.9 to 5.9 GHz; at the 5.64 GHz operating frequency, a single element provides approximately 6 dBi gain, 86% radiation efficiency, and a half-power beamwidth of approximately 70°. Each RF chain comprises a variable attenuator and phase shifter to set the complex excitation, followed by a per-channel Skyworks SE5004L-R power amplifier (PA). The same PA device is operated with either a 5 or 3.3 V supply in the two active states. A dual-channel power-management integrated circuit (PMIC) generates the 5 and 3.3 V supply rails, and the detailed beamforming-board supply architecture is shown in Fig. 3. In the active states, the 5 V rail supplies the non-PA circuitry of the beamforming board, including the control and RF-control blocks, while the PA supply voltage $V_{DD,PA}$ is selected as either 5 or 3.3 V through the EN1/EN2 enable-control, inverter, and switch circuitry. Before entering the OFF state, a microcontroller unit (MCU) sets the variable attenuator to its maximum-attenuation state and de-asserts the PA enable/reference input V_{REF} . Both PMIC output channels are then disabled, thereby disconnecting $V_{DD,PA}$. Consequently, although the common upstream RF feed may remain present, the RF drive delivered to the PA is minimized and the unpowered, disabled PA provides no active amplification. The rail-selection variables therefore represent three hardware states for each transmit chain: active with a 5 V PA supply, active with a 3.3 V PA supply, or board-level OFF. Here, the optimized excitation vector is generated on a personal computer (PC) and transferred to the MCU through universal asynchronous receiver/transmitter (UART) communication. The MCU then configures the phase shifter and variable attenuator through serial peripheral interface (SPI)-based digital control and sets the enable states of the PMIC/switching path to realize the selected 5 and 3.3 V, or OFF PA state. The radiated RF power propagates through free space to the receiver, where an RF power splitter feeds multiple rectifiers. This parallel rectifier architecture maintains operation near each rectifier's high-efficiency region while delivering a higher combined DC output. Before introducing the proposed hardware-aware objective, the conventional RF-domain MISO WPT baseline is written as a received-RF-power-constrained

transmit-output-power minimization problem, as commonly used in beamforming-based WPT formulations [10], [11]

$$\begin{aligned} \min_{A_i} \quad & \sum_{i=1}^M |A_i|^2 \\ \text{s.t.} \quad & \left| \sum_{i=1}^M A_i B_i \right|^2 \geq P_r. \end{aligned} \quad (1)$$

In (1), a MISO WPT system with M transmit elements and a single receive antenna is assumed, as illustrated in Fig. 2. To clarify the received power expression, the excitation of the i th transmit element is written as $A_i e^{j\phi_i}$, where A_i and ϕ_i denote the excitation amplitude and phase, respectively. The corresponding complex channel coefficient from the i th transmit element to the receive antenna is written as $B_i e^{j\theta_i}$, where B_i represents the channel magnitude and θ_i represents the channel phase. The effective channel magnitude coefficient is defined as follows:

$$B_i = \sqrt{G_t(\theta_{t,i}, \phi_{t,i}) G_r(\theta_{r,i}, \phi_{r,i})} \frac{\lambda_0}{4\pi d_i} \quad (2)$$

where λ_0 is the free-space wavelength, d_i is the propagation distance between the i th transmitting element and the receiving antenna, and G_t and G_r denote the transmitting and receiving antenna gains in the corresponding directions, respectively. Therefore, B_i includes the element-dependent propagation distance as well as the directional antenna-gain terms, and it is used as the deterministic channel-magnitude coefficient for the fixed measurement geometry. Thus, the field contribution from the i th element at the receiver is given by $A_i B_i e^{j(\phi_i + \theta_i)}$, and the total received field is obtained by coherently summing these contributions over all transmit elements. The received RF power can therefore be expressed as $P_r = \left| \sum_{i=1}^M A_i B_i e^{j(\phi_i + \theta_i)} \right|^2$. Because a single receive antenna is considered in the MISO case, the transmit phase can be directly aligned with the channel phase by setting $\phi_i = -\theta_i$. After this phase alignment, the received power constraint is expressed only in terms of the nonnegative amplitudes A_i and the channel magnitudes B_i , as shown in (1). In contrast to this conventional transmit power minimization (i.e., SDR method-based convex optimization), the proposed formulation constrains the received power while minimizing transmitter hardware power consumption, as expressed in

$$\begin{aligned} \min_{A_i} \quad & \sum_{i=1}^M f(A_i) \\ \text{s.t.} \quad & \left| \sum_{i=1}^M A_i B_i \right|^2 \geq P_r \end{aligned} \quad (3)$$

where P_r is the target received RF power selected from the measured RF-to-DC conversion characteristics of the receiver rectifier. In this work, P_r is not treated as an additional continuous optimization variable; instead, it is fixed as a rectifier-aware design target so that the receiver operates near its measured peak efficiency region. The detailed determination of P_r for the implemented receiver, including the rectifier peak efficiency point, is described in the Appendix. Thus,

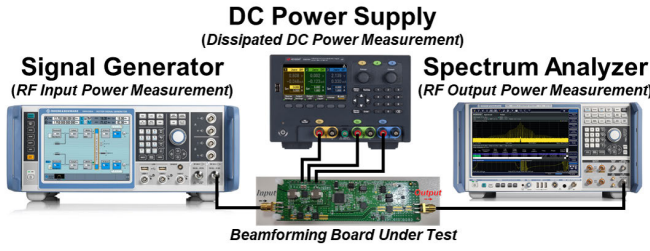


Fig. 4. Measurement environment of the proposed beamforming board for the results of the dissipated power and RF power.

receiver characteristics are incorporated through the choice of P_r , while the proposed optimization focuses on minimizing the transmitter-side hardware power consumption under this received-power requirement. This constrained formulation was chosen instead of directly maximizing an efficiency ratio because the measured PA and board-level efficiency characteristics are generally nonconvex with respect to output power, especially across different supply rails. In contrast, the measured transmitter DC-dissipation function can be represented by a convex regression model over the operating range, which enables a tractable global optimization framework. The advantage of this approach is that the receiver-side power requirement and rectifier operating point are guaranteed, whereas the limitation is that P_r must be selected according to the intended load and receiver characterization. Furthermore, here, unlike the conventional method of minimizing formulations for the radiated RF power, this objective explicitly captures transmitter hardware information (i.e., DC power consumption of the beamforming board); coupled with the second-order constraint, global optimality is guaranteed when $f(\cdot)$ is *convex*. In our approach, $f(A_i)$ is obtained via regression based on measured power consumption data. We enforce convexity by applying convexity constraints to piecewise linear (PL) functions. To propose an appropriate functional form, we use a reference function defined by a polynomial regression without convexity constraints. The derivation is presented sequentially below. Additionally, here, A_i is expressed in the field-amplitude, or voltage-equivalent, domain rather than directly in the power domain. This representation is used because the received fields from different transmit elements are coherently summed before the received RF power is obtained by taking the squared magnitude.

B. Regression of the Transmitter Power Consumption Function

To characterize the power dissipation and derive a convex regression model, measurement-based data were acquired using the setup shown in Fig. 4. The fabricated beamforming board was supplied by a DC power supply, from which the board-level consumed DC power was obtained. This measured power includes the PA, PMIC, MCU, phase shifter, variable attenuator, and other RF/control components on the beamforming board. A single-tone RF input at the operating frequency was applied using a signal generator, and the RF output power was measured using a spectrum analyzer.

By sweeping the output amplitude under each supply condition, measured data pairs between the output amplitude and the corresponding DC power consumption were obtained. Here, the output amplitude denotes the RF voltage-equivalent amplitude converted from the measured RF output power in the 50- Ω system, not the DC supply voltage. These data were then used to construct the amplitude-dependent dissipation models for the 5 and 3.3 V supply rails. The present characterization therefore focuses on single-tone amplitude-dependent power dissipation; broadband or multitone operation would require additional characterization of linearity and harmonic behavior over the relevant bandwidth. Thus, the PA and board-level nonlinearities relevant to this single-tone operation are not ignored, but are embedded in the measured amplitude-to-DC-dissipation data before the convex regression is applied. The resulting per-channel dissipation models are applied to the identically implemented array channels, and their aggregate validity under simultaneous full-array operation is evaluated through the measurement results in Section IV.

The measured data are then converted into a scalar dissipation function because the optimization in (3) is formulated with respect to the transmit amplitude A_i . In this regression, x_i denotes the i th measured RF output amplitude sample, $y_{data,i}$ denotes the corresponding measured DC power consumption, and $y_{fit,i}$ denotes the fit dissipation value to be determined at the same amplitude sample. The vectors $\mathbf{y}_{data} = [y_{data,1}, \dots, y_{data,N}]^T$ and $\mathbf{y}_{fit} = [y_{fit,1}, \dots, y_{fit,N}]^T$ collect the measured and fit dissipation samples over all N amplitude points, respectively. The goal of the regression is therefore to obtain an amplitude-dependent function that maps A_i to the transmitter DC power dissipation. The convexity of this fit function is important because the objective in (3) is given by the sum of the element-wise dissipation functions. If each dissipation model is convex, the resulting amplitude optimization preserves the convex structure. For the proposed PL regression, this convexity is enforced by constraining the segment slopes to be nondecreasing over the measured amplitude range. The constrained regression problem is formulated as follows [21]:

$$\begin{aligned} \min_{\mathbf{y}_{fit}} \quad & \sum_{i=1}^N (y_{fit,i} - y_{data,i})^2 \\ \text{s.t.} \quad & \frac{y_{fit,i+1} - y_{fit,i}}{x_{i+1} - x_i} \geq \frac{y_{fit,i} - y_{fit,i-1}}{x_i - x_{i-1}} \\ & (\forall i = 2, \dots, N-1). \end{aligned} \quad (4)$$

In (4), the objective minimizes the squared fitting error between the measured dissipation samples and the fit dissipation samples. The constraint enforces nondecreasing slopes (i.e., the derivative of the function) between adjacent amplitude intervals, which is the convexity condition for the PL dissipation model. This constrained least squares problem can be equivalently written as the following quadratic programming (QP) formulation [9]:

$$\begin{aligned} \min_{\mathbf{y}_{fit}} \quad & \frac{1}{2} \mathbf{y}_{fit}^T (2I) \mathbf{y}_{fit} + (-\mathbf{y}_{data})^T \mathbf{y}_{fit} \\ \text{s.t.} \quad & A_{ineq,PL} \mathbf{y}_{fit} \leq \mathbf{b}_{PL}, \quad \mathbf{b}_{PL} = \mathbf{0}. \end{aligned} \quad (5)$$

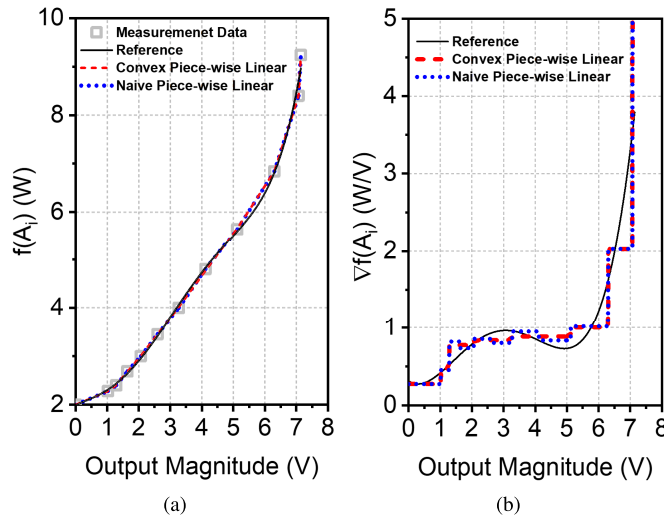


Fig. 5. Measurement-based regression of the beamforming board for the 5 V supply voltage. (a) Measured board-level DC power dissipation versus RF output voltage-equivalent amplitude and fitted dissipation models. (b) Derivative comparison of the fit models.

The inequality matrix is defined as follows:

$$[A_{ineq,PL}]_{(i-1),j} = \begin{cases} \frac{1}{x_i - x_{i-1}}, & \text{if } j = i - 1 \\ \frac{x_i - x_{i-1}}{(x_i - x_{i-1})(x_{i+1} - x_i)}, & \text{if } j = i \\ \frac{1}{x_{i+1} - x_i}, & \text{if } j = i + 1 \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

Here, $\mathbf{b}_{PL}$ is the right-hand side vector of the linear inequality representation. It becomes the zero vector because the nondecreasing slope condition in (4) can be rearranged into a homogeneous linear inequality with all terms involving $\mathbf{y}_{fit}$ moved to the left-hand side.

Finally, by solving the minimization problem in (4), we obtain the PL function

$$f_{PL} = y_{fit,i} + \frac{y_{fit,i+1} - y_{fit,i}}{x_{i+1} - x_i} (x - x_i)$$

for $x \in [x_i, x_{i+1}]$ under convexity constraints. Now, the function $f(A_i)$ is identified for the 5 V supply voltage; its regression is shown in Fig. 5(a) with gray square markers indicating measurement data. For comparison, the conventional fifth-order polynomial fit (reference), naive PL fit, and proposed convex PL regression were plotted. Although the fits look similar in $f(\cdot)$, the derivative plot in Fig. 5(b) reveals that the reference and naive PL exhibit nonmonotonic slopes (i.e., $\partial f / \partial A$ is not increasing), violating convexity. Such behavior can induce local optima in the optimization. In contrast, the convex PL regression enforces monotonicity and convexity, preserving the global optimality guarantees of the convex optimization. The same procedure is applied to the supply voltage of 3.3 V to obtain $g(A_i)$, which is used together with $f(A_i)$ in the final formulation.

For reference, the polynomial order used for the unconstrained smooth trend function was selected by comparing polynomial orders 5, 6, and 7, as shown in Fig. 6. The

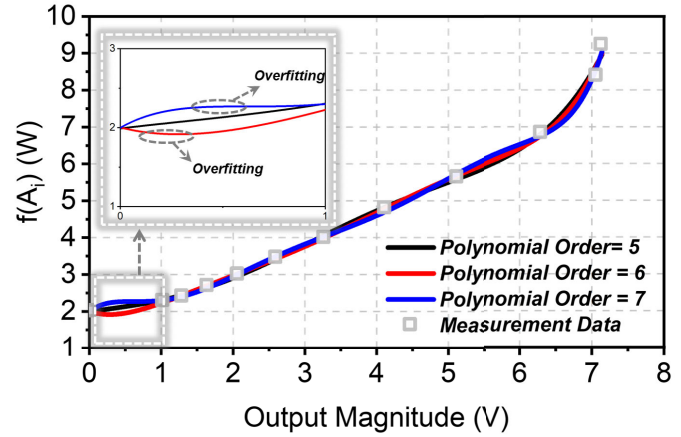


Fig. 6. Polynomial order comparison as a reference function for the measurement-based regression.

reference polynomial is not used as the final dissipation model in the optimization; rather, it is introduced only to provide a smooth baseline for interpreting the measured power consumption trend. From a physical viewpoint, the consumed DC power of the beamforming board should not locally decrease as the output magnitude increases within the measured operating range. As shown in Fig. 6, the sixth-order polynomial exhibits a nonmonotonic region near the low output magnitude range, where the fit consumed power locally decreases despite an increase in output magnitude. The seventh-order polynomial shows an even stronger local distortion, including an abrupt increase followed by a slight decrease in the same low amplitude region. Because these intersample variations are neither supported by the measured samples nor consistent with the expected monotonic power-consumption trend, they are interpreted as overfitting of the higher-order unconstrained polynomials in the present regression context. Therefore, the fifth-order polynomial was selected as the highest-order candidate that follows the measured samples without introducing unsupported intersample oscillations. It should also be noted that ordinary polynomial fitting alone does not guarantee monotonicity or convexity, even when the overall fit curve appears to follow the measured data. This is because the unconstrained least squares fit minimizes the global fitting error but does not impose physical constraints on the derivative or curvature. For this reason, the polynomial fit is used only as a reference function, while the final transmitter dissipation model used in the optimization is obtained from the convex PL regression with nondecreasing slope constraints. The derivative of each fit model was therefore evaluated to verify whether the fit dissipation model satisfies the required convexity condition. The derivations of the reference polynomial and the other regression methods for comparison are presented in the Appendix.

Additionally, the analysis presented thus far has focused on the 5 V supply voltage. The corresponding results for the 3.3 V supply voltage, which was introduced as an additional state for supply voltage switching, are also shown in Fig. 7(a) and (b). Accordingly, the regression for $g(A_i)$ was carried out using the same procedure so that the model fits the measurement data while maintaining convexity.

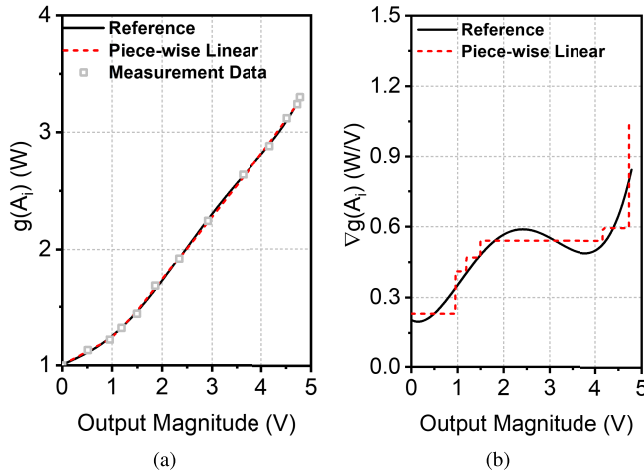


Fig. 7. Measurement-based regression of the beamforming board for the 3.3 V supply voltage. (a) Measured board-level DC power dissipation versus RF output voltage-equivalent amplitude and fit dissipation models. (b) Derivative comparison of the fit models.

III. HARDWARE-AWARE VOLTAGE SWITCHING AS A MIXED-INTEGER LINEAR PROGRAM

The proposed hardware-aware optimization is developed in four steps. First, the convex base formulation in Section II is extended to a 2-state mixed-integer linear programming (MILP) that jointly determines the transmit amplitudes and the 5/3.3 V rail assignment. Second, the formulation is relaxed to a 3-state MILP by allowing the OFF state through the rail-selection inequality. Third, dead-index embedding and slack-penalized relaxation are introduced to handle faulty elements and potentially infeasible received-power targets. Finally, a soft-transition-based warm start is constructed to initialize the binary rail-selection variables and reduce the branch-and-bound burden. For convenience, the principal symbols, functions, and auxiliary variables used throughout Sections II and III are summarized in the Nomenclature.

We now extend the convex formulation in (3) to explicitly model *rail selection* at each element among {5, 3.3V} for 2-state operation and {5, 3.3V, OFF} for 3-state operation. Here, a rail refers to the supply state provided to the transmitter chain, and rail selection means choosing whether each element operates with the 5 V supply, the 3.3 V supply, or is turned off. Let $A_{i,5} \geq 0$ and $A_{i,3.3} \geq 0$ denote the RF output amplitude contributions of the i th element when the 5 and 3.3 V rails are selected, respectively, and let $A_i = A_{i,5} + A_{i,3.3}$. These amplitude variables correspond to the magnitude level converted from the measured RF output power of the beamforming board. In the measurement, the RF input power was swept under each supply condition, and the corresponding RF output power and DC power consumption were recorded to construct the rail-specific amplitude and dissipation models. The upper bounds $A_{\max,5}$ and $A_{\max,3.3}$ represent the maximum available output amplitudes under the 5 and 3.3 V rails, respectively, and are determined by the measured output capability of the transmitter chain, including the PA saturation limit for each supply state. We introduce binary variables $z_{i,5}, z_{i,3.3} \in \{0, 1\}$

to indicate which rail is used; for the 2-state model we enforce $z_{i,5} + z_{i,3.3} = 1$, whereas for the 3-state model $z_{i,5} + z_{i,3.3} \leq 1$ so that *OFF* is allowed. Let $\mathcal{D}$ be the set of dead elements.

A. 2-State Model (5 V or 3.3 V)

Starting from the continuous base formulation in (3), we derive a 2-state MILP that jointly determines the transmit amplitudes and the 5/3.3 V rail assignment for each alive element. This formulation is constructed in this work by combining the measured convex PL dissipation models with binary rail-selection variables, epigraph constraints, and dead-index embedding

$$\begin{aligned}
 \min_{\substack{A_i, t_{i,5}, t_{i,3.3} \geq 0, \\ z_{i,5} \in \{0,1\}, s \geq 0}} \quad & \sum_{i=1}^M (t_{i,5} + t_{i,3.3}) + \mu s \\
 \text{s.t.} \quad & \sum_{i=1}^M A_i B_i + s \geq \sqrt{P_r} \\
 & A_i = A_{i,5} + A_{i,3.3}, \quad 0 \leq A_i \leq A_{\max} \\
 & 0 \leq A_{i,5} \leq A_{\max,5} z_{i,5} \\
 & 0 \leq A_{i,3.3} \leq A_{\max,3.3} z_{i,3.3} \\
 & A_{i,5} = 0, t_{i,5} = 0, z_{i,5} = 0 \quad \forall i \in \mathcal{D} \\
 & z_{i,5} + z_{i,3.3} = 1 \quad \forall i \notin \mathcal{D} \\
 & t_{i,5} \geq m_{\ell}^{(f)} A_{i,5} + c_{\ell}^{(f)} - M_5 z_{i,5} \quad \forall \ell \\
 & t_{i,3.3} \geq m_{\ell}^{(g)} A_{i,3.3} + c_{\ell}^{(g)} - M_{3.3} z_{i,3.3} \quad \forall \ell \\
 & t_{i,5} \leq T_5^{\max} z_{i,5}, t_{i,3.3} \leq T_{3.3}^{\max} z_{i,3.3} \quad (7)
 \end{aligned}$$

where $Z_{i,v} = (1 - z_{i,v})$, $A_{\max} = \max\{A_{\max,5}, A_{\max,3.3}\}$, and $\mathcal{D}$ indexes dead elements. Here, $t_{i,5}$ and $t_{i,3.3}$ are epigraph variables that upper bound the transmitter dissipation associated with the 5 and 3.3 V rails, respectively. The convex PL rail-wise dissipation models are represented by affine segments of the form $m_{\ell}^{(f)} A + c_{\ell}^{(f)}$ for the 5 V rail and $m_{\ell}^{(g)} A + c_{\ell}^{(g)}$ for the 3.3 V rail, where $m_{\ell}^{(f)}$ and $c_{\ell}^{(f)}$ denote the slope and intercept of the ℓ th segment, respectively. In addition, $T_v^{\max}$ denotes an upper bound on the corresponding epigraph variable $t_{i,v}$ over the feasible amplitude range of rail v , and is used to deactivate the epigraph term through the tight link $t_{i,v} \leq T_v^{\max} z_{i,v}$ when that rail is not selected.

B. 3-State Model (5/3.3 V/OFF)

Allowing OFF is obtained by relaxing the exclusivity equality to an inequality

$$\begin{aligned}
 \min \quad & \sum_{i=1}^M (t_{i,5} + t_{i,3.3}) + \mu s \\
 \text{s.t.} \quad & \text{constraints in (7) except } z_{i,5} + z_{i,3.3} = 1 \\
 & \text{replace it with } z_{i,5} + z_{i,3.3} \leq 1 \quad \forall i \notin \mathcal{D}. \quad (8)
 \end{aligned}$$

This formulation captures the discrete rail selection with power gating while keeping the objective convex in the continuous variables.

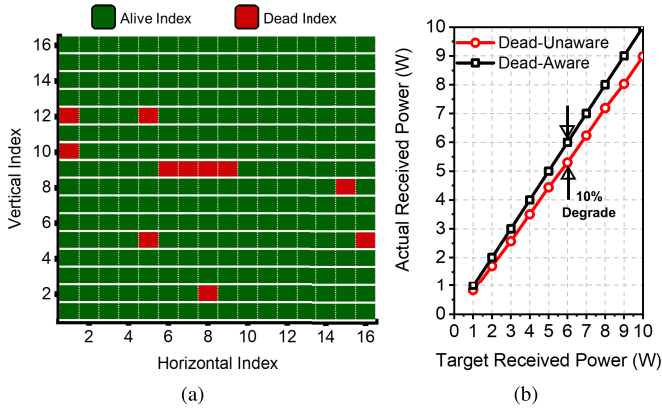


Fig. 8. (a) 16×16 array transmitter including dead index. (b) Actual received power comparison of MILP for the hardware-aware.

C. Dead-Index Embedding

In this work, a dead element refers to a malfunctioning transmit element whose RF response cannot be properly controlled during the calibration process. The implemented array was calibrated using the free-space channel compensation procedure in [5], where the element-wise amplitude and phase states were characterized by controlling the attenuator and phase shifter of each beamforming board channel. During this process, some elements showed no meaningful response even when their amplitude or phase control states were changed. These nonresponsive elements were identified as dead elements and collected in the dead index set $\mathcal{D}$.

Dead elements $i \in \mathcal{D}$ are handled by fixing the corresponding variables to zero, i.e.,

$$A_{i,\cdot} = 0, \quad t_{i,\cdot} = 0, \quad z_{i,\cdot} = 0.$$

This approach directly reflects the hardware incompleteness in the optimization model without introducing additional integer variables. Such a consideration is particularly important in large-scale array systems. During practical operation, some elements may become faulty or may stop radiating because of hardware breakdown, so that their amplitude and phase can no longer be properly controlled. Although replacement of these elements would be possible in principle, this is not always practical in large-scale systems where maintenance accessibility and scalability vary substantially. In such situations, it is often more efficient to preserve the hardware as is and incorporate the faulty elements into the optimization problem itself. This is the motivation for introducing the dead index in the present work.

Fig. 8(a) shows the actual status of the implemented array, where alive elements are marked in green and nonoperating elements are marked in red. Fig. 8(b) compares the cases where the dead indices are either considered or ignored. When the dead elements are ignored in the optimization model, the optimizer assumes that all array elements can contribute to the coherent received field. As a result, nonzero excitation may be assigned to elements that are physically nonresponsive in the implemented hardware. During the actual measurement, however, these dead elements do not radiate the intended amplitude and phase contribution, so the realized coherent

sum at the receiver becomes smaller than the optimized target. This model hardware mismatch causes the actual received power to become approximately 10% lower than the intended value. This degradation not only reduces the received power itself, but also shifts the operating point away from the peak efficiency region of the rectifier, thereby degrading the overall system efficiency. These results demonstrate that the dead index treatment is essential for maintaining both the validity of the optimization result and the intended end-to-end performance of the WPT system.

D. Slack Penalized Feasibility Relaxation

The slack variable $s \geq 0$ in (7) and (8) is introduced to ensure that the optimization problem remains solvable even when the target received power P_r cannot be achieved because of limited transmit amplitude or the presence of dead elements. In this sense, the slack provides a best-effort solution in infeasible cases. On the other hand, when the target is feasible and the penalty coefficient μ is sufficiently large, the optimal solution satisfies $s^* = 0$, so that the received power constraint is met without relaxation.

To clarify the role of the slack term, the received power constraint is written in the square root form as follows:

$$\sum_{i=1}^M A_i B_i + s \geq \sqrt{P_r} \quad (9)$$

where B_i is the effective channel magnitude coefficient defined in Section II-A. It includes the element-dependent propagation distance and the transmit and receive antenna gain terms, and therefore represents the magnitude weighting with which the excitation of the i th transmit element contributes to the coherent received field. Thus, B_i is used in the square root received power constraint as the effective contribution coefficient of each transmit element, rather than as a pure path attenuation term.

Since dead elements do not participate in power transfer, it is convenient to define

$$\mathbf{B}_{\text{alive}} \triangleq [B_i]_{i \notin \mathcal{D}} \quad (10)$$

where $\mathcal{D}$ is the dead index set. In other words, $\mathbf{B}_{\text{alive}}$ is the subvector composed only of the effective channel magnitude coefficients corresponding to alive elements.

Using $\mathbf{B}_{\text{alive}}$, a representative transmit amplitude scale is defined as follows:

$$A_{\text{scale}} \triangleq \frac{\sqrt{P_r}}{\|\mathbf{B}_{\text{alive}}\|_2}. \quad (11)$$

This quantity may be interpreted as the characteristic norm level of the transmit amplitude vector required to satisfy the target $\sqrt{P_r}$ when the excitation is aligned with the effective channel. Hence, A_{scale} serves as a convenient normalization factor for the received power requirement.

A representative dissipation scale is also introduced from the epigraph upper bounds of the rail-wise dissipation variables. Recall that $t_{i,v}$ denotes the epigraph variable associated with the dissipation on rail v , and $T_v^{\max}$ is its feasible upper bound over the corresponding amplitude range. Since $T_5^{\max}$ and $T_{3,3}^{\max}$

denote the maximum epigraph values for the 5 and 3.3 V rails, respectively, we define

$$T_{\text{scale}} \triangleq \max \{T_5^{\max}, T_{3.3}^{\max}\}. \quad (12)$$

This provides a conservative upper scale for the element-wise hardware dissipation.

Based on these quantities, the penalty coefficient μ may be selected using either of the following scale-aware rules:

$$\mu \geq \eta \cdot \frac{m_{\max}}{B_{\max}} \quad \text{or} \quad \mu = \eta \cdot \frac{T_{\text{scale}}}{A_{\text{scale}}} \quad (13)$$

where

$$m_{\max} = \max \left\{ \max_{\ell} m_{\ell}^{(f)}, \max_{\ell} m_{\ell}^{(g)} \right\}, \quad B_{\max} = \max_{i \notin \mathcal{D}} B_i \quad (14)$$

and $\eta \in [10, 100]$ is a safety factor; here, $m_{\ell}^{(f)}$ and $m_{\ell}^{(g)}$ denote the slopes of the ℓ th affine segment in the convex PL dissipation models $f(\cdot)$ and $g(\cdot)$, respectively. Therefore, $m_{\max}$ represents the largest marginal increase in dissipation with respect to amplitude among all PL segments across the two supply rails.

The physical meaning of these rules is as follows. The first rule compares the largest slope of the rail-dependent dissipation model with the strongest available effective channel magnitude coefficient. Therefore, one unit increase in the slack is penalized more heavily than the marginal reduction in objective that could be obtained by relaxing the received power requirement. The second rule compares a representative dissipation scale with a representative amplitude scale, thereby providing a more intuitive normalization-based choice for μ .

In both cases, the objective is the same: the penalty on the slack variable should dominate the benefit of violating the target received power constraint. As a result, the optimizer uses $s > 0$ only when the requested P_r is genuinely infeasible because of hardware limitations such as insufficient amplitude range or dead elements.

In the present implementation, μ was chosen as a sufficiently large constant so that the slack remained inactive for feasible targets. Nevertheless, the scale-aware rules in (13) provide a practical guideline for selecting μ in a systematic manner for other operating conditions or array configurations.

E. Warm Start via Soft Transition

To reduce the computational burden of branch-and-bound, a continuous warm start is generated before solving the final MILP. The main idea is to first construct a relaxed solution for the 2-state problem, and then extend it to the 3-state case by additionally introducing an OFF decision for sufficiently small amplitudes.

Before the epigraph reformulation, the 2-state switching problem may be interpreted as selecting either the 5 V rail or the 3.3 V rail for each alive element. In this case, the objective can be written as follows:

$$\min_{\mathbf{A}, \mathbf{z}} \sum_{i=1}^M (z_{i,5} f(A_i) + z_{i,3.3} g(A_i)) \quad (15)$$

subject to the received power constraint and

$$z_{i,5} + z_{i,3.3} = 1 \quad \forall i \notin \mathcal{D} \quad (16)$$

where $z_{i,5}$ and $z_{i,3.3}$ are binary variables indicating the selected rail. However, directly using binary variables in the initialization stage is inconvenient. Therefore, the discrete rail selection is replaced by a continuous variable $w_i \in [0, 1]$, where $w_i \approx 0$ corresponds to the 5 V rail and $w_i \approx 1$ corresponds to the 3.3 V rail.

Using this relaxation, the element-wise surrogate objective is defined as follows:

$$h_i(A_i, w_i) = (1 - w_i) f(A_i) + w_i g(A_i). \quad (17)$$

Accordingly, the warm start is obtained by minimizing $\sum_i h_i$ while satisfying the amplitude range and the received power constraint. Since phase conjugation is already imposed in the MISO setting, the received power constraint is expressed in the linearized form (9)

The relaxed rail preference is updated from the difference between the two rail-wise dissipation models

$$\Delta_i = g(A_i) - f(A_i). \quad (18)$$

If $\Delta_i < 0$, the 3.3 V rail is locally more efficient; if $\Delta_i > 0$, the 5 V rail is preferred. To make this transition smooth and differentiable, a sigmoid function is used

$$w_i^{\text{new}} = \frac{1}{1 + \exp(\Delta_i/\beta)} \quad (19)$$

where β controls the softness of the transition. In addition, when $|\Delta_i|$ is very small, the rail decision may become unstable near the switching boundary. To avoid excessive oscillation, the previous value is retained in that region, and the update is damped as follows:

$$w_i \leftarrow (1 - \rho) w_i + \rho w_i^{\text{new}} \quad (20)$$

where ρ is a relaxation factor.

A further hardware-dependent correction is required because the 3.3 V rail cannot support amplitudes larger than $A_{\max,3.3}$. Therefore, whenever

$$A_i > A_{\max,3.3} \quad (21)$$

the corresponding relaxed variable is forced toward the 5 V side. In other words, the soft transition already incorporates the feasible operating range of the 3.3 V rail.

Once w_i is updated, the amplitude vector is refined by a projected gradient step using the surrogate objective in (17). The resulting warm start pair is denoted by $(\mathbf{A}^{\text{soft}}, \mathbf{w}^{\text{soft}})$.

The next step is to convert this relaxed solution into the initial rail assignment for the 2-state MILP. For clarity, the superscript “init” is used to denote the initial value provided to the MILP solver. The initial rail selection is assigned as follows:

$$z_{i,3.3}^{\text{init}} = \begin{cases} 1, & \text{if } w_i^{\text{soft}} > 0.5 \\ 0, & \text{otherwise} \end{cases} \quad (22)$$

$$z_{i,5}^{\text{init}} = \begin{cases} 1, & \text{if } w_i^{\text{soft}} \leq 0.5 \\ 0, & \text{otherwise} \end{cases} \quad (23)$$

Thus, the relaxed variable w_i^{soft} is simply thresholded at 0.5 to determine whether the element starts from the 5 V rail or the

3.3 V rail. Based on this binary initialization, the continuous variables are set consistently as follows:

$$A_{i,5}^{\text{init}} = \min(A_i^{\text{soft}}, A_{\max,5}) z_{i,5}^{\text{init}} \quad (24)$$

$$A_{i,3.3}^{\text{init}} = \min(A_i^{\text{soft}}, A_{\max,3.3}) z_{i,3.3}^{\text{init}} \quad (25)$$

and

$$A_i^{\text{init}} = A_{i,5}^{\text{init}} + A_{i,3.3}^{\text{init}}. \quad (26)$$

Finally, the initial slack is set as follows:

$$s^{\text{init}} = \max \left\{ 0, \sqrt{P_r} - \sum_{i=1}^M A_i^{\text{init}} B_i \right\}. \quad (27)$$

The above procedure directly provides the warm start for the 2-state MILP. The 3-state warm start is then obtained by extending the same logic. In the final 3-state formulation, the equality constraint in (16) is relaxed to

$$z_{i,5} + z_{i,3.3} \leq 1 \quad \forall i \notin \mathcal{D} \quad (28)$$

which allows the OFF state. However, this change alone is not sufficient in the warm start stage. An additional rule is needed to determine which elements should be initialized as OFF.

To achieve this, the same thresholding on w_i^{soft} is first applied to determine whether the active rail should be 5 or 3.3 V. Then, among the elements that are not assigned to 3.3 V, those with negligibly small relaxed amplitudes are initialized as OFF. Specifically, in the 3-state case

$$z_{i,3.3}^{\text{init}} = \begin{cases} 1, & \text{if } w_i^{\text{soft}} > 0.5 \\ 0, & \text{otherwise} \end{cases} \quad (29)$$

and

$$z_{i,5}^{\text{init}} = \begin{cases} 1, & \text{if } w_i^{\text{soft}} \leq 0.5 \text{ and } A_i^{\text{soft}} > \varepsilon_{\text{off}} \\ 0, & \text{otherwise} \end{cases} \quad (30)$$

where ε_{off} is a small threshold used to identify amplitudes that are effectively zero. Therefore, if an element is not assigned to the 3.3 V rail and its relaxed amplitude is sufficiently small, it is initialized as OFF. This additional OFF thresholding is the main difference between the 2-state and 3-state warm start procedures.

As a result, the 2-state warm start determines the initial rail assignment by thresholding the relaxed variable w_i^{soft} , whereas the 3-state warm start further refines this result by identifying low-amplitude elements that should start from the OFF state. All dead elements are fixed to zero throughout the procedure. In this way, the warm start remains consistent with the rail-dependent dissipation models, the amplitude capability of the 3.3 V rail, and the structure of the final MILP, thereby significantly reducing the solve time.

IV. IMPLEMENTATION AND EXPERIMENTAL RESULTS

To isolate the algorithmic contribution from differences in hardware and receiver configuration, Fig. 9 presents a controlled same-platform comparison of transmitter-side dissipated power. The transmitting array, calibrated channel coefficients, receiver configuration, amplitude limits, measured 5/3.3 V dissipation models, and target received RF power sweep are identical in all four cases; only the optimization

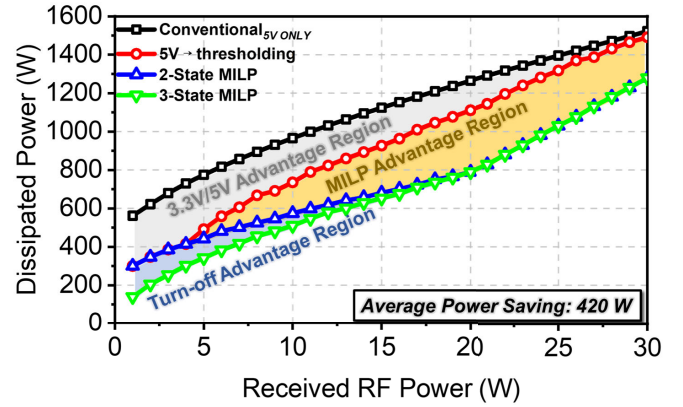


Fig. 9. Comprehensive comparison of transmitter-side dissipated power versus target received RF power for four methods: 5 V only, 5 V → thresholding, 2-state MILP, and 3-state MILP.

objective and the treatment of the rail and OFF states are changed. The four cases are the conventional 5 V-only formulation that minimizes transmit RF output power (black), heuristic rail thresholding applied after the conventional RF excitation is determined (red), the 2-state MILP that jointly optimizes the amplitudes and the {3.3, 5 V} rail states (blue), and the proposed 3-state MILP that additionally allows the OFF state (green). Relative to the conventional 5 V-only baseline, the proposed 3-state MILP reduces the transmitter-side dissipated power by approximately 420 W on average and by up to approximately 60%.

The horizontal axis in Fig. 9 denotes the target received RF power used in the parametric optimization sweep, not the received DC power. The sweep up to 30 W is used to examine the operating regions in which rail switching, MILP-based rail assignment, and OFF-state operation become effective over the transmitter operating range; the experimental demonstration was performed at the 2 W received-RF target selected from the receiver characteristics.

Beyond the overall ranking of the four methods, the relative gap between the curves provides additional physical insight into the role of each design element. First, the gap between the conventional black curve and the red thresholding curve indicates the gain that can already be obtained by introducing rail switching between 3.3 and 5 V, even without embedding the switching decision into the optimization itself. For this reason, the corresponding operating range is denoted as the 3.3/5 V advantage region. This result shows that a noticeable reduction in dissipated power can be achieved solely by allowing the hardware to operate with multiple supply voltages, even when the beamforming amplitudes are still derived from the conventional objective.

Second, the relationship between the red thresholding curve and the blue 2-state MILP curve reveals the benefit of jointly optimizing the beamforming amplitudes together with the rail assignment. In the low received power region, the two curves nearly coincide. This behavior indicates that, under weak received power constraints, the optimal solution of the 2-state MILP converges to the 3.3 V-only state, so that the naive thresholding rule and the embedded optimization lead to

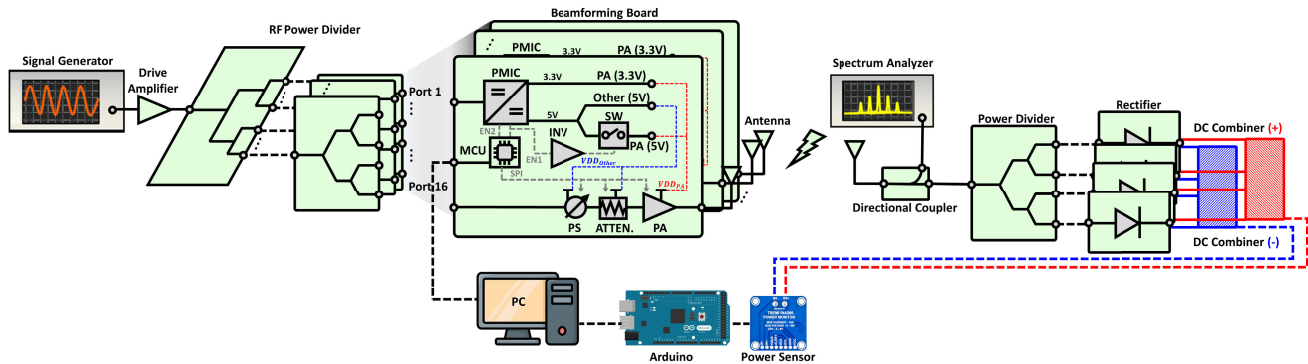


Fig. 10. Block diagram of the experimental validation setup. The control data are transferred from the PC to the MCU through a USB-to-UART interface, and the MCU configures the beamforming boards through SPI/control lines. The received RF power is monitored through a directional coupler, while the main RF path is divided into four rectifier branches and the combined DC output is measured through the power-sensor interface.

nearly the same operating point. As the target received power increases, however, the two curves gradually separate. In this regime, the required transmit amplitudes can no longer be supported efficiently by a single rail, and the optimal solution involves a nontrivial mixture of 3.3 and 5 V states across the array. Therefore, the blue curve achieves a lower dissipated power than the red curve because the rail selection is no longer determined by a postprocessing rule, but is instead cooptimized with the amplitude distribution inside the MILP. This operating range is thus denoted as the MILP advantage region.

Third, the gap between the blue 2-state MILP curve and the green 3-state MILP curve quantifies the benefit of introducing the turn-off state. This gap is most pronounced in the low target received power region and gradually decreases as the required received power becomes larger. Such behavior is also physically consistent. When the received power requirement is low, many array elements are not required to contribute significantly to the coherent sum, and it becomes beneficial to turn them off completely rather than keeping them active at a low amplitude. In contrast, as the received power constraint becomes stronger, a larger portion of the array must remain active, so the opportunity to exploit the turn-off state is reduced. In the high received power regime, the 3-state MILP therefore approaches the 2-state MILP, because the solution is dominated by active 3.3 and 5 V operation rather than by power gating. For this reason, the low-power operating range where the green curve maintains an additional benefit is referred to as the turn-off advantage region.

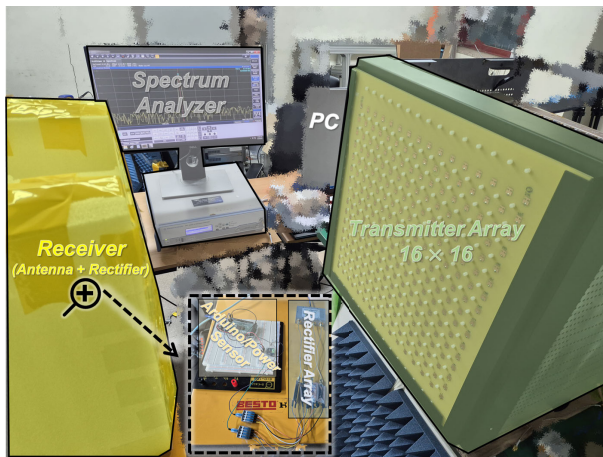
Taken together, these comparisons provide more than a sequential demonstration of performance improvement. They also confirm that the proposed modeling and implementation steps are physically well grounded. The 3.3/5 V advantage region verifies the benefit of multirail operation itself, the MILP advantage region confirms the necessity of embedding the rail decision into the optimization, and the turn-off advantage region validates the additional gain enabled by explicit power gating. Therefore, the measured ordering of the four methods is not merely numerical, but is fully consistent with the expected hardware behavior of the transmitter. This agreement between optimization result and physical interpretation

strongly supports both the validity of the proposed formulation and the practicality of the implemented WPT system.

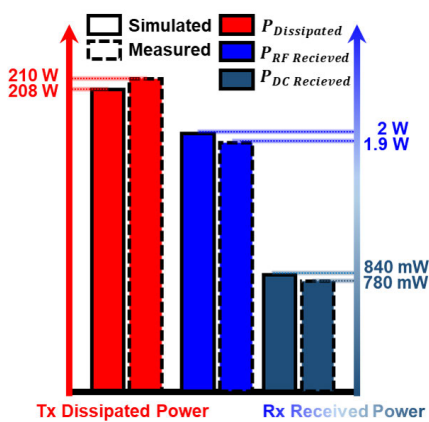
Fig. 10 shows the block diagram of the experimental validation setup, including the transmitter control path, RF power-distribution path, and receiver-side DC measurement path. On the transmitter RF path, the signal-generator output is amplified by a driver amplifier and distributed to the 256 beamforming-board channels through two cascaded 16-way RF power-divider stages. The driver amplifier provides a fixed common drive level to compensate for the RF power-divider loss, whereas the per-channel PAs realize the optimized amplitude and supply-state operation. The driver-amplifier consumption is included as a fixed term in the total transmitter-power budget and therefore does not affect the MILP solution. The optimized excitation and rail-state data are generated on a PC and transferred to the MCU through a USB-to-UART interface, after which the MCU configures the phase shifters, variable attenuators, and supply-state controls through SPI/control lines.

To validate the proposed optimization framework, the measurement setup shown in Fig. 11(a) was implemented in a MISO configuration at a separation distance of 0.5 m. The transmitter employed a 16×16 array with 256 antenna elements, while the receiver used a single antenna element of the same type. On the receiver side, the received RF power was monitored through the coupled port of a directional coupler using a spectrum analyzer, while the main RF path was delivered to a four-way power divider and distributed to four rectifier branches. After rectification, the four DC outputs were combined and measured through the power-sensor interface connected to the PC via an Arduino board.

The target received RF power was selected according to the rectifier operating characteristics described in the Appendix. Since each rectifier branch achieved its peak efficiency around an input RF power of 400 mW, corresponding to approximately 210 mW DC output, the receiver was designed to use four parallel rectifier branches with a target total DC output of about 840 mW. Considering the insertion loss of the four-way power divider, the received RF power target of the MISO WPT system was set to 2 W, and the transmitter excitation was optimized accordingly.



(a)



(b)

Fig. 11. (a) Measurement environment. (b) Results of simulated and measured dissipated transmitter power, received RF and DC power at the receiver at 0.5 m.

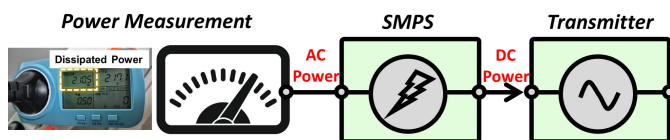


Fig. 12. Transmitter input-power measurement setup with the wall-plug input power measured before the SMPS.

The measured results are presented in Fig. 11(b). In Fig. 11(b), the simulated transmitter power denotes the estimated value obtained from the optimized hardware-realizable amplitude, phase, and rail-state set, including the fixed driver-amplifier consumption, whereas the measured transmitter power denotes the power directly measured at the wall-plug input of the transmitter power-supply system. Therefore, the measured value includes the SMPS conversion loss and provides a conservative transmitter-input-power reference for system-level efficiency evaluation, as shown in Fig. 12. The transmitter consumed 210 W, the receiver obtained 1.9 W of RF power, and the rectifier produced 780 mW of DC power. Accordingly, the measured receiver-module RF-to-DC conversion efficiency was 41.1%. Using the directly measured

transmitter input power as the denominator, the transmitter input-to-RF efficiency was $29/210 \text{ W} = 13.8\%$, including the practical power-supply conversion loss. The single-branch rectifier characteristic used to select the target received power is provided in the Appendix, where the rectifier operates near its peak efficiency around 400 mW input RF power and 210 mW DC output. Compared with the target or simulated values shown in Fig. 11(b), the measured transmitter input power, received RF power, and received DC power differ by approximately 1.0%, 5.0%, and 7.1%, respectively. Because the complete-array measurement inherently includes residual mutual-coupling and active-impedance effects, the close agreement between the target or model-predicted values and the measured transmitter input power and received RF power indicates that these effects are not dominant for the implemented array at the tested operating point. This agreement supports the practical validity of applying the per-channel dissipation model to the implemented full-array system.

Table I summarizes a system-level comparison with recently reported WPT systems by separating the power flow into three reference-plane efficiencies. The transmitter DC-to-RF efficiency, $\eta_{\text{txDC} \rightarrow \text{txRF}}$, mainly reflects the transmitter hardware, including the PA efficiency, supply-voltage condition, and power-management architecture. The transmitted-RF-to-received-RF efficiency, $\eta_{\text{txRF} \rightarrow \text{rxRF}}$, corresponds to the over-the-air power-transfer efficiency and is strongly affected by the propagation distance, aperture size, channel condition, and Tx/Rx configuration. In particular, this metric can vary significantly depending on whether the system uses a MISO or multiple-input multiple-output (MIMO) receiver architecture, and is also the metric most directly affected by beamforming, TR, or optimization-based excitation design. The received RF-to-DC efficiency, $\eta_{\text{rxRF} \rightarrow \text{rxDC}}$, is mainly determined by the rectifier topology, input power level, and load condition. The total end-to-end efficiency is defined as the ratio of the received DC output to the transmitter-side input or consumed power. This value is reported only when both endpoint quantities are available under the same operating condition; otherwise, it is denoted as N.R. to avoid unsupported estimation.

Because these efficiency components are governed by different hardware blocks and system configurations, Table I should not be interpreted as a simple ranking based on a single numerical value. The reported systems differ in the number of receive elements, propagation distance, receiver architecture, physical aperture, RF-chain implementation, and rectifier design. Therefore, the comparison is intended to clarify how each system distributes its performance across the transmitter, propagation channel, and receiver, rather than to provide a strictly controlled one-to-one benchmark. For this reason, the table also includes the algorithm method, qualitative computational burden, P_{Diss} , handling, and transmitter aperture size. The computational-burden entry indicates the dominant complexity of determining the transmit excitation, while the transmitter aperture size refers only to the physical antenna-array aperture.

At the same time, the comparison can still be used to identify method-level differences when systems with equivalent

TABLE I
COMPARISON WITH RECENTLY PUBLISHED WPT SYSTEMS

Metric	[5]	[14]	[22]	[23]	[24]	[25]	This work
System Configuration							
Algorithm method	Convex Opt.	Convex Opt. + heuristic switching	Beamforming	Time Reversal	Beamforming	Retrodirective	3-state MILP
Config. (Tx × Rx)	256 × 35	256 × 25	9 × 1	64 × 1	16 × 16	16 × 2	256 × 1
Distance (λ_0^{**})	9.4	9.4	8	69	77	4	9.4
Power Metrics							
Tx RF power (W)	6.08	13.6 ^b	11.7	32 ^c	1.3	3.98 ^d	29
Received RF power (W)	4.37	10.6	0.37	0.16 ^c	0.007	> 0.50 ^d	1.9
Tx dc power (W)	241	237	48.6 ^a	N.R.	N.R.	N.R.	210
Received dc power (W)	N.R.	N.R.	N.R.	0.10 ^c	N.R.	> 0.35 ^d	0.78
Efficiency Metrics							
$\eta_{\text{txDC} \rightarrow \text{txRF}}$ (%)	2.52	5.73 ^b	24.1 ^a	N.R.	N.R.	N.R.	13.8
$\eta_{\text{txRF} \rightarrow \text{rxRF}}$ (%)	71.9	78.0 ^b	3.2	0.50 ^c	0.54	> 12.6 ^d	6.55
$\eta_{\text{rxRF} \rightarrow \text{rxDC}}$ (%)	N.R.	N.R.	N.R.	62.5 ^c	N.R.	> 70 ^d	41.1
Total end-to-end efficiency (%)	N.R.	N.R.	N.R.	N.R.	N.R.	N.R.	0.37
Algorithmic and Physical Footprint							
$P_{\text{Diss.}}$ handling	No	Heur.	No	No	No	No	Opt. obj.
Computational burden	High	High	Low	Low	Low	Low	Med.^g –High
Tx aperture size (m × m)	0.50 × 0.50	0.50 × 0.50	N.R.	~0.23 × 0.23 ^e	~0.10 × 0.10 ^e	~0.25 × 0.32 ^f	0.50 × 0.50

$P_{\text{tx,dc}}$ denotes the transmitter-side dissipated power (wall-plug measurement) used in this comparison. The efficiency metrics are defined as $\eta_{\text{txDC} \rightarrow \text{txRF}} = 100P_{\text{tx,RF}}/P_{\text{tx,dc}}$, $\eta_{\text{txRF} \rightarrow \text{rxRF}} = 100P_{\text{rx,RF}}/P_{\text{tx,RF}}$, and $\eta_{\text{rxRF} \rightarrow \text{rxDC}} = 100P_{\text{rx,dc}}/P_{\text{rx,RF}}$.

In the $P_{\text{Diss.}}$ handling row, “No” means that transmitter-side dc/dissipated power is not included in the optimization or control objective, “Heur.” means that it is considered through heuristic switching or post-processing, and “Opt. obj.” means that it is directly embedded in the optimization objective.

N.R.: not reported under a directly comparable operating condition. For the total end-to-end efficiency, N.R. is used when either the transmitter-side input or consumed power or the received dc output is unavailable at the same operating point.

^a The transmitter-side dc/dissipated power is inferred from the reported PA efficiency.

^b The transmit RF power is back-calculated from the reported received RF power and the reported free-space efficiency of 78%.

^c The total transmitted RF power and received dc power are taken from the reported demonstration; the RF-to-dc efficiency is calculated using the received RF power retained in this comparison.

^d The RF powers in [25] are converted from the reported dBm values, and the corresponding efficiencies are shown using the reported 27-dBm received-RF reference.

^e The Tx aperture size is approximately estimated from the reported Tx array topology and operating frequency by assuming a uniform half-wavelength element pitch, i.e., $D_x \times D_y \approx N_x \lambda_0/2 \times N_y \lambda_0/2$. The value denotes an estimated antenna-aperture footprint.

^f The Tx aperture size of [25] is approximately estimated from the reported 4-subarray, 16-element transmitter geometry.

^g Medium in the practical MISO implementation sense. The proposed formulation includes binary rail-selection variables, but SDR is avoided by direct MISO phase alignment and the MILP is initialized using the proposed warm-start procedure. Therefore, the practical computational burden is moderate, while the worst-case complexity remains higher than deterministic beamforming or TR-based methods.

** Free-space wavelength at the operating frequency.

or closely related operating assumptions are considered. For example, Yoon et al. [14] use a related large-array transmitter platform and consider transmitter consumption through

heuristic supply-voltage switching; however, its higher received power and over-the-air efficiency are also aided by an MIMO receiver configuration. Thus, for a fairer

MISO-level interpretation, this approach is closer to the “5 V $\rightarrow$ thresholding” case in Fig. 9, where the rail state is selected heuristically after the conventional excitation is determined. Under this equivalent interpretation, the proposed MILP-based method exhibits lower transmitter-side dissipated power because the beamforming amplitudes and rail states are jointly optimized. Similarly, [5] employs a related MIMO WPT system but does not explicitly model transmitter-side dissipation in the optimization; therefore, its method is closer in spirit to the “5V-only” baseline in Fig. 9. Taken together, Table I provides system-level context across heterogeneous WPT platforms, whereas Fig. 9 provides the primary algorithm-level benchmark because the transmitter hardware, calibrated channel, receiver configuration, amplitude constraints, and received RF power requirement are held fixed. The correspondence of the cited prior approaches to the 5 V-only and heuristic-thresholding baselines is methodological and is not intended as a numerical reproduction of those reported systems.

For the implemented 256×1 MISO demonstration, the measured values $P_{\text{tx,DC}} = 210$ W for the directly measured transmitter input power, $P_{\text{tx,RF}} = 29$ W, $P_{\text{rx,RF}} = 1.9$ W, and $P_{\text{rx,DC}} = 0.78$ W correspond to $\eta_{\text{txDC} \rightarrow \text{txRF}} = 13.8\%$, $\eta_{\text{txRF} \rightarrow \text{rxRF}} = 6.55\%$, and $\eta_{\text{txRF} \rightarrow \text{rxDC}} = 41.1\%$, respectively. These values should be understood together with the controlled comparisons in Fig. 9: the proposed method is not claimed to dominate every subsystem metric independently, but rather to improve the transmitter-side consumed power when the hardware state, supply-voltage selection, and turn-off operation are incorporated into the optimization itself.

V. DISCUSSION ON THE PROPOSED HARDWARE-AWARE OPTIMIZATION FRAMEWORK

This section discusses the implications of the proposed hardware-aware formulation beyond the numerical comparison itself. In particular, the discussion aims to clarify why the proposed approach should not be regarded as a mere extension of the conventional formulation, but rather as a necessary reformulation when the transmitter hardware characteristics and supply voltage switching are explicitly taken into account.

A. Comparison With Conventional Output Power Optimization in the 5 V-Only Single-Rail Case

The first observation is obtained from the comparison between the conventional output power minimization and the proposed dissipated power minimization under the 5 V-only condition, as shown in Fig. 13. In this comparison, supply-voltage switching and OFF-state operation are intentionally disabled so that both formulations operate under the same single-rail transmitter environment. Therefore, Fig. 13 isolates the effect of replacing the conventional transmit-output-power objective with the proposed transmitter-dissipated-power objective. Under this condition, the result shows that minimizing the conventional transmit output power yields a solution that is not significantly different from that obtained by directly minimizing the dissipated DC power.

More specifically, the difference between the two solutions is approximately 5%, with the proposed dissipated power

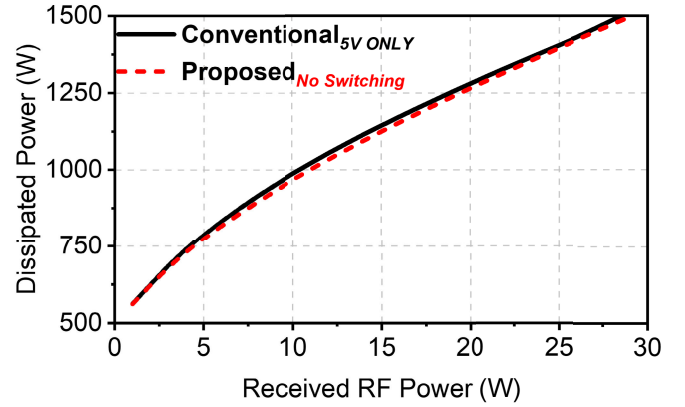


Fig. 13. Single-rail comparison of transmitter-side dissipated DC power versus target received RF power under the 5 V-only condition. The conventional method minimizes transmit RF output power, whereas the proposed method minimizes the measured transmitter-dissipation model, with supply-voltage switching and OFF-state operation disabled in both cases.

optimization requiring about 5% less consumed power to satisfy the same target received power. This result indicates that, when only a single supply voltage is used, the conventional formulation already provides a reasonably meaningful solution. The physical reason is that the output power of the PA and the corresponding dissipated power still maintain a strong monotonic relationship, even though the relationship is not perfectly linear. As a result, minimizing the transmit output power indirectly follows the trend of minimizing the actual consumed power.

This observation is important because it verifies that the conventional method is not fundamentally invalid in a single-rail environment. On the contrary, the result confirms that the conventional output power-based optimization can serve as a meaningful baseline when the transmitter hardware is constrained to operate only at 5 V. In this sense, the proposed formulation does not merely outperform an artificially weak reference, but is instead compared against a baseline that remains physically relevant and practically useful in the absence of supply voltage switching.

At the same time, this result also provides a key implication for the remainder of the discussion. Since the gap between the two formulations is relatively small in the single-rail case, the dramatic performance improvement observed after introducing supply voltage switching cannot be explained solely by a minor refinement of the same objective. Rather, it indicates that once multiple hardware states are introduced, the conventional output power objective is no longer sufficient to represent the actual transmitter-side efficiency. Therefore, the need for the proposed dissipated-power-based formulation becomes more evident in the multistate setting, where dissipated power is no longer well represented by output power alone.

B. Why Supply Voltage Switching Requires Dissipated Power Optimization

A more important implication emerges when the final 3-state supply voltage switching result is compared with the two no-switching cases, as shown in Fig. 14. In this figure, the

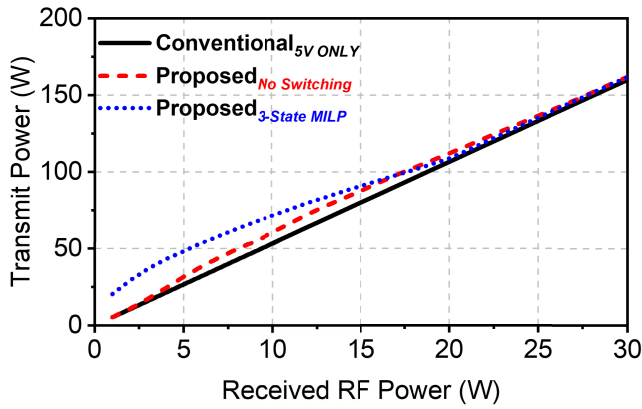


Fig. 14. Results comparison of output power for conventional and proposed methods with 5 V only supply voltage and proposed 3-state MILP.

transmit output power obtained from the conventional formulation, the proposed dissipated power optimization under the 5 V only condition, and the final 3-state MILP result are compared on the same basis. As expected, the conventional output power minimization yields the lowest transmit output power over most of the operating range. This is a natural consequence of its objective, since the formulation is constructed to reduce the radiated transmit power itself.

However, this result also reveals the key limitation of the conventional viewpoint. Although the conventional solution produces the lowest output power, it does not lead to the lowest transmitter-side consumed power once supply voltage switching is introduced. On the contrary, the proposed 3-state MILP occasionally results in a higher transmit output power than the no-switching solutions, while still achieving a significantly lower consumed DC power. As already shown in Fig. 9, the proposed switching strategy reduces the dissipated power by approximately 420 W on average, and by about 60% in the low received power regime. This clearly indicates that, in the presence of multiple supply states, the transmit output power alone is no longer an adequate proxy for the actual hardware efficiency.

The physical reason is that supply voltage switching changes the relationship between output power and consumed power in a fundamentally discrete and state-dependent manner. In the single-rail case, output power minimization can approximately follow dissipated power minimization because both quantities remain strongly correlated through the monotonic behavior of the PA. Once multiple supply voltages and the OFF state are introduced, however, the same output amplitude may correspond to very different levels of hardware dissipation depending on which rail is selected. Consequently, a solution with slightly larger output power may still be substantially more efficient if it enables operation in a more favorable supply state or turns off inefficient elements altogether.

Therefore, Fig. 14 should not be interpreted merely as a comparison of transmit power levels. Rather, it demonstrates that the optimization objective itself must be changed when supply voltage switching is incorporated into the hardware. In such a multistate transmitter environment, minimizing output power is insufficient to identify the true power-efficient operating point. This is precisely why the proposed

dissipated power-based formulation is required. The result confirms that the proposed framework, consisting of a convex base formulation and an MILP-based switching extension, is not simply an incremental extension of the conventional approach, but a necessary reformulation for achieving genuine end-to-end transmitter efficiency improvement under supply voltage switching.

VI. CONCLUSION

This article proposed a hardware-aware convex optimization framework for a 5.64 GHz MISO WPT system, in which transmitter-side power dissipation was modeled directly from measured beamforming board data and incorporated into the optimization objective. Unlike conventional formulations that minimize only transmit RF power, the proposed approach employed convex PL regression for the 5 and 3.3 V supply rails and embedded supply voltage switching into MILP formulations for both 2-state and 3-state operation. To improve practical applicability, the framework was further extended with slack-penalized feasibility relaxation, soft-transition-based warm-start generation, and dead-index embedding so that infeasible targets, discrete rail assignment, and faulty array elements could all be treated systematically within the optimization.

The experimental results obtained with a 16×16 transmitting array and a single receiving antenna at 0.5 m verified the effectiveness of the proposed method. The implemented WPT system achieved 1.9 W of received RF power and 780 mW of DC output while drawing 210 W of directly measured transmitter input power, including the SMPS conversion loss. This consumed power is distinguished from the transmitted RF power, which was 29 W in the implemented demonstration. In addition, the comparison among conventional optimization, naive thresholding, 2-state MILP, and 3-state MILP confirmed that the proposed 3-state formulation provides the largest reduction in dissipated power, with an average saving of approximately 420 W.

Beyond the numerical improvement itself, the proposed work also clarified an important physical insight: in a 5 V-only single-rail environment, conventional output-power minimization can still provide a meaningful baseline because output power and dissipated power remain strongly correlated. However, once multiple supply states and turn-off operation are introduced, output power alone is no longer an adequate proxy for hardware efficiency. Therefore, the proposed dissipated-power-based optimization framework, built on a convex base formulation and extended through MILP-based supply voltage switching, is not merely an incremental extension of conventional WPT optimization, but a necessary reformulation for realizing genuine end-to-end efficiency improvement under realistic hardware conditions.

APPENDIX

A. Summary of Notation

Nomenclature provides a compact reference for the principal notation used in the system model, dissipation regression, MILP formulation, slack relaxation, and warm-start procedure.

Appendix-only local coefficients are defined where they first appear.

B. Additional Regression Candidates and Reference Model Selection

Although the main text adopts the convex PL regression as the final dissipation model, additional regression candidates were also investigated to validate this choice. The purpose of this section is therefore twofold. First, a smooth reference curve is introduced to provide a qualitative baseline for the measured dissipation trend. Second, several convex regression candidates are compared against this reference to justify the final use of the PL model in the proposed optimization framework. In particular, this section considers three convex candidates, namely convex PL, convex piecewise quadratic (PQ), and convex polynomial (P) regression, while the reference model is chosen as an unconstrained fifth-order polynomial. This role of the section is consistent with the main text, where the transmitter dissipation model is ultimately implemented using convex PL regression, whereas the polynomial fit is used only as a reference for comparison.

1) *Reference Regression Using an Unconstrained Fifth Order Polynomial*: To obtain a smooth baseline that captures the overall trend of the measured dissipation data, the reference function was chosen as a fifth-order polynomial

$$f_{\text{ref}}(x) = \sum_{k=0}^5 a_k x^k. \quad (31)$$

The polynomial coefficients are determined by the ordinary least squares problem

$$\min_{\mathbf{a}} \sum_{i=1}^N (f_{\text{ref}}(x_i) - y_{\text{data},i})^2 \quad (32)$$

where x_i denotes the measured output amplitude and $y_{\text{data},i}$ is the corresponding measured power dissipation. The fifth order was selected as a compromise between fitting flexibility and physical consistency. In particular, higher order polynomials were avoided because they tend to introduce artificial oscillations and physically unreasonable behavior, such as a locally decreasing dissipation trend as the output magnitude increases. This order selection is used only for the smooth reference curve; convexity is evaluated separately through the derivative profile and is enforced in the final convex PL model.

2) *Convex PQ Regression*: As a first alternative to the final PL model, a convex PQ regression was considered. Let

$$f_{PQ,i}(x) = a_i x^2 + b_i x + c_i, \quad x \in [x_i, x_{i+1}] \quad (33)$$

denote the quadratic model on the i th interval. The coefficients are obtained from

$$\begin{aligned} \min_{a_i, b_i, c_i} \quad & \sum_{i=1}^{N-1} \sum_{x_j \in [x_i, x_{i+1}]} (f_{PQ,i}(x_j) - y_{\text{data},j})^2 \\ \text{s.t.} \quad & a_i \geq 0 \\ & f_{PQ,i}(x_{i+1}) = f_{PQ,i+1}(x_{i+1}) \\ & f'_{PQ,i}(x_{i+1}) = f'_{PQ,i+1}(x_{i+1}) \end{aligned} \quad (34)$$

where $a_i \geq 0$ enforces convexity on each segment, while the equality constraints guarantee continuity and differentiability at the interval boundaries.

By collecting the unknown coefficients into

$$\boldsymbol{\theta} = [a_1, b_1, c_1, \dots, a_{N-1}, b_{N-1}, c_{N-1}]^T \quad (35)$$

the problem can be recast as the quadratic program

$$\begin{aligned} \min_{\boldsymbol{\theta}} \quad & \frac{1}{2} \boldsymbol{\theta}^T (2O^T O) \boldsymbol{\theta} + (-2O^T \mathbf{y})^T \boldsymbol{\theta} \\ \text{s.t.} \quad & A_{\text{ineq},PQ} \boldsymbol{\theta} \leq \mathbf{0} \\ & A_{\text{eq},PQ} \boldsymbol{\theta} = \mathbf{0} \end{aligned} \quad (36)$$

where $\mathbf{y}$ denotes the measured dissipation samples and O is the corresponding design matrix for the PQ basis.

The inequality matrix for segment-wise convexity is given by

$$[A_{\text{ineq},PQ}]_{i,j} = \begin{cases} -1, & j = 3(i-1) + 1 \\ 0, & \text{otherwise} \end{cases} \quad (37)$$

which is equivalent to enforcing $a_i \geq 0$. The equality matrix $A_{\text{eq},PQ}$ contains two rows per interface to impose continuity and first-derivative continuity. Specifically, the rows associated with function continuity are

$$[A_{\text{eq},PQ}]_{\text{even row},j} = \begin{cases} x_{i+1}^2, & j = 3(i-1) + 1 \\ x_{i+1}, & j = 3(i-1) + 2 \\ 1, & j = 3(i-1) + 3 \\ -x_{i+1}^2, & j = 3i + 1 \\ -x_{i+1}, & j = 3i + 2 \\ -1, & j = 3i + 3 \\ 0, & \text{otherwise} \end{cases} \quad (38)$$

and the rows associated with derivative continuity are

$$[A_{\text{eq},PQ}]_{\text{odd row},j} = \begin{cases} 2x_{i+1}, & j = 3(i-1) + 1 \\ 1, & j = 3(i-1) + 2 \\ -2x_{i+1}, & j = 3i + 1 \\ -1, & j = 3i + 2 \\ 0, & \text{otherwise.} \end{cases} \quad (39)$$

3) *Convex Polynomial Regression*: As another candidate, a single global polynomial with convexity constraints was also considered

$$\begin{aligned} \min_{\mathbf{a}} \quad & \sum_{i=1}^N (f_P(x_i; \mathbf{a}) - y_{\text{data},i})^2 \\ \text{s.t.} \quad & f_P''(x) \geq 0, \quad x \in [x_{\min}, x_{\max}] \end{aligned} \quad (40)$$

where

$$f_P(x) = \sum_{k=0}^p a_k x^k. \quad (41)$$

This can be written as the quadratic program

$$\begin{aligned} \min_{\mathbf{a}} \quad & \frac{1}{2} \mathbf{a}^T (2X^T X) \mathbf{a} + (-2X^T \mathbf{y})^T \mathbf{a} \\ \text{s.t.} \quad & A_{\text{ineq},P} \mathbf{a} \leq \mathbf{0} \end{aligned} \quad (42)$$

TABLE II
MSE AND MAE COMPARISON AMONG CONVEX
REGRESSION CANDIDATES

	MSE (mW) [5V/3.3V]	MAE (mW) [5V/3.3V]
PL	5.5 / 0.177	54.8 / 11.1
PQ	5.7 / 0.178	59.2 / 11.5
P	10.2 / 0.301	80.1 / 15.1

where the Vandermonde-type matrix is

$$X = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^p \\ 1 & x_2 & x_2^2 & \cdots & x_2^p \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_N & x_N^2 & \cdots & x_N^p \end{bmatrix} \quad (43)$$

and the inequality matrix is constructed from the second derivative condition as follows:

$$[A_{ineq,P}]_{i,j} = \begin{cases} -k(k-1)x^{k-2}, & j = k+1, \quad k = 2, \dots, p \\ 0, & \text{otherwise.} \end{cases} \quad (44)$$

4) *Model Comparison and Final Selection:* Based on the above derivations, three convex regression candidates were prepared: PL, PQ, and convex polynomial regression. These candidates were not compared directly against the raw measured data alone, but also against the reference fifth-order polynomial in (31), which serves as a smooth baseline describing the overall dissipation trend. The comparison metrics were the mean-squared error (mse) and mean absolute error (MAE), defined as follows:

$$\text{MSE} = \frac{1}{K} \sum_{i=1}^K (y_{\text{ref},i} - y_{\text{fit},i})^2 \quad (45)$$

$$\text{MAE} = \frac{1}{K} \sum_{i=1}^K |y_{\text{ref},i} - y_{\text{fit},i}| \quad (46)$$

where K is the number of densely sampled points over the range $x \in [x_{\min}, x_{\max}]$.

Table II summarizes the comparison. Among the three convex candidates, the PL model exhibited the smallest mse and MAE for both the 5 and 3.3 V supply cases. This result justifies the final choice made in the main text: although smoother alternatives such as PQ and convex polynomial regression are possible, the convex PL model provides the most accurate approximation to the reference trend while maintaining a simple structure that is directly compatible with the epigraph-based MILP formulation. Therefore, the Appendix confirms that the final use of PL regression in the proposed hardware-aware optimization is not arbitrary, but is supported by both accuracy and implementation considerations.

C. Rectifier Design and Target Received Power

The target received RF power was determined based on the operating characteristics of the rectifier used in the measurement setup. The rectifier was designed using a voltage doubler topology and implemented with HSMS-282C diodes [26].

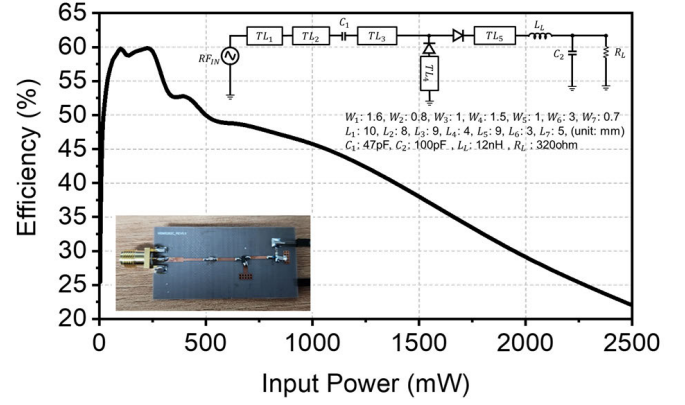


Fig. 15. Measured result and schematic of the rectifier in the proposed WPT system.

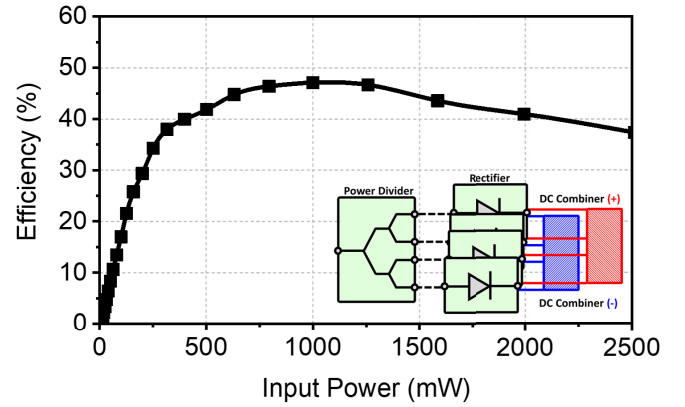


Fig. 16. Measured module-level RF-to-DC performance of the four-branch receiver, including the four-way RF power divider.

Its measured performance at the operating frequency of 5.64 GHz is shown in Fig. 15, where the RF-to-DC conversion efficiency is evaluated as a function of the input RF power. As observed in Fig. 15, the single rectifier branch maintains high-efficiency operation around an input RF power of approximately 400 mW. At this operating point, the rectifier delivers about 210 mW of DC power. Based on this result, the proposed receiver was configured using four rectifier branches so that a high combined DC output can be obtained while each branch operates near its intended high-power, high-efficiency region.

In practice, the received RF power is first distributed to the four rectifier branches through a four-way power divider. To verify this practical receiver configuration, the complete four-branch receiver module was measured, as shown in Fig. 16. In this measurement, the input RF power is defined at the receiver-module input before the four-way divider, and the measured efficiency therefore includes the divider insertion loss, RF cable loss, branch-to-branch variation, rectifier mismatch, and DC combining effect.

Compared with the single-branch result, the module-level response is shifted by approximately 6 dB along the input-power axis because the input RF power is divided into four branches. The module-level efficiency is also lower and broader than the single-branch efficiency because the

complete receiver measurement includes the practical RF division and combining losses. Around the 2 W receiver-module input region, the measured module-level efficiency is about 41%, which agrees well with the WPT demonstration result of 1.9 W received RF power and 780 mW combined DC output.

This rectifier-aware power target is consistent with the system-level design philosophy of the proposed work. Rather than selecting the received RF power arbitrarily, the optimization target was chosen so that the complete receiver module operates in a high-efficiency region while maintaining a sufficiently high combined DC output. Consequently, the measured WPT demonstration was performed near the 2 W received RF power target, resulting in 780 mW combined DC output.

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